Zhan Jin

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JustinJin04.github.io

Education

University of Illinois Urbana-Champaign, Champaign, IL *Aug 2026 - present*
Ph.D. in Computer Science

- Advised by Prof. Francis Y. Yan

Peking University, Beijing, China *Sep 2022 - June 2026*
BS in Computer Science

- Outstanding Graduate of the Class of 2026 in the major of Computer Science and Technology, School of EECS

Research Interests

I'm pursuing two directions: building efficient and scalable systems that enable advances in foundation model capabilities, while designing algorithms whose benefits increase monotonically with model capabilities.

Publications

Accelerating Cross-Model LLM Inference via Attention Map Reuse
Zhan Jin, Yuze Luo, Bin Cui, Xupeng Miao.
SIGMOD 2027

Tactic: Adaptive Sparse Attention with Clustering and Distribution Fitting for Long-Context LLMs
Kan Zhu, Tian Tang, Qinyu Xu, **Zhan Jin**, Yile Gu, Zhichen Zeng, Rohan Kadekodi, Liangyu Zhao, Ang Li, Arvind Krishnamurthy, Baris Kasikci.
ICLR 2026

Preprints

DualTune: Decoupled Fine-tuning for On-Device Agentic Systems
Rohan Kadekodi*, **Zhan Jin***, Keisuke Kamahori, Yile Gu, Sean Khatiri, Noah H. Bayindirli, Sergey Gorbunov, Baris Kasikci.
*Equal contribution

Research Experience

Data and Intelligence Research (DAIR) Lab, PKU *Sep 2025 - Apr 2026*
Advisor: Professor Xupeng Miao, Bin Cui
Project: Accelerating Cross-Model LLM Inference via Attention Map Reuse

- Designed and evaluated cross-model attention-map reuse, enabling same-tokenizer models of different sizes to reach near-lossless accuracy across multiple benchmarks.
- Developing an efficient end-to-end runtime by implementing optimized kernels and compressing stored attention maps, providing 15 to 20 % lower TTFT and up to 25 % higher throughput.

Systems for Future Intelligence (SyFI) Lab, UW *Jun 2025 - Sep 2025*
Advisor: Professor Baris Kasikci
Project: Decoupled Fine-tuning for On-Device Agentic Systems

- Analyzed the performance and accuracy constraints in current on-device, tool-calling agent frameworks.
- Introduced a modular approach that separates tool selection and argument generation, identifying tool selection as the dominant error source.
- Proposed decoupled fine-tuning with specialized adapters for each subtask, improving tool-calling accuracy by +46% on MCP-Bench compared to the base 7B model.

Awards

- **Outstanding Graduate of the Class of 2026 in the major of Computer Science and Technology, School of EECS, 2026**
- **Second Prize, Integrated Circuit EDA Design Elite Challenge, Chinese Institute of Electronics, 2023**
- **Academic Excellence Award, Peking University, 2024**
- **Third Prize, Peking University Scholarship, 2024**
- **Outstanding Research Award, Peking University, 2025**

Skills/Interests

- Extensive programming, solid expertise on C, C++, CUDA, Python
- Familiar with distributed training systems (Megatron-LM, DeepSpeed) and inference systems (vLLM, SGLang)
- Interested in Linux-based software development and embedded systems.

Activities

- **Volunteer:** Volunteer teacher in Midu, Yunnan (Chinese once national poverty county)
- **Sports:** Passionate about baseball, basketball, soccer, badminton, swimming, and hiking.